Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/21/2025
PI Name:	Levine, Benjamin D M.D.		
Project Title:	Coronary Anatomy and Physiology During 1 Year in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	214-345-4619
Organization Name:	The University of Texas Southwestern Medical Center at Dallas		
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PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75231-5129	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2019 HERO 80JSC018N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	04/20/2020	**End Date:**	04/19/2034
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 4/19/2034 per HHC element and NSSC information (Ed., 6/21/21)		
Key Personnel Changes/Previous PI:	PI Benjamin D. Levine, MD ; CoI Michael W. Bungo, MD ; CoI Jonathan R. Lindner, MD [Note Linda Loerch is no longer CoI];		
COI Name (Institution):	Bungo, Michael M.D. (University of Texas Health Science Center at Houston) Lindner, Jonathan R. M.D. (University of Virginia Medical Center)		
Grant/Contract No.:	80NSSC20K0987		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Since the majority of experienced astronauts are middle aged, they are at risk for developing serious cardiovascular events such as a myocardial infarction or sudden cardiac death, especially during high intensity exertion. Studies led to the current flight medicine practice of screening all astronaut candidates (and following all active crew members) with coronary artery calcium (CAC) scoring. However, atherosclerosis is a progressive process. The development of vascular calcification may be preceded by substantial non-calcified plaque, which may be most prone to rupture and cause an acute coronary syndrome. Radiation and inflammation may exacerbate this natural history. Coronary atherosclerosis impairs coronary endothelial function which can then both initiate and stimulate progression of atherosclerosis. Recent flight studies have suggested that non-coronary vascular beds may stiffen with reduced vascular reserve during 6-month International Space Station (ISS) missions, and ground-based studies have identified the surprising capacity for coronary atherosclerosis to evolve rapidly under extreme stress. In addition, the Principal Investigator (PI) team recently completed the Integrated CardioVascular (ICV) study that demonstrated: a) although cardiac arrhythmias did not increase in space in most astronauts, unexpectedly, left atrial (LA) size increased out of proportion to the changes in left ventricular (LV) size; and b) there was a subset of crew (1/13 or 8%) who had substantial increases in both ventricular and atrial arrhythmias. These data raise the specter of increased risk for atrial fibrillation (AF), the most common arrhythmia in the US which occurs a decade earlier in astronauts than in the general population. We speculate that the combination of spaceflight plus exercise countermeasures could magnify LA dilation and lead to AF during a 2-3 yr Mars mission. AF in astronauts is a particular concern with prolonged spaceflight because of limited access to care and the risk of impaired exercise performance, poorly controlled ventricular response, deterioration of ventricular function, and arterial emboli (including stroke).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Learning more about the natural progression of atherosclerosis in the spaceflight environment may have "spin-off" benefits for characterizing these processes in terrestrial populations. In addition, out exploratory aim in this project is to determine if a blood biomarker panel might be predictive of alterations in the atherosclerotic process. Should this prove useful, the direct benefit to clinical care on Earth would be significant.
Task Progress:	Experiments designed to accomplish these goals continue as crewmembers volunteer, are assigned a flight, and complete the proposed studies. It is too early to generate meaningful conclusions. However, the team has met, reviewed all completed imaging studies, and are happy with the data quality and results. We continue to recruit new crewmembers to volunteer for our CIPHER experiments as crews are identified and Informed Consent Briefings (ICBs) are scheduled. [Ed. Note: CIPHER is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). CIPHER stands for "Complement of Integrated Protocols for Human Exploration Research".] Two subjects have completed pre-flight studies and will fly during the next reporting period. Two other subjects will undergo pre-flight studies in the next year, and one will complete their 1-year follow up studies. All data are analyzed immediately, checked for quality, and periodically reviewed by the entire team.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
Articles in Peer-reviewed Journals	MacNamara JP, Hearon CM, Manferdelli G, Shah AM, Tayon KG, Pandey A, Satyam S, Levine BD. "Heart failure in zero gravity-external constraint and cardiac hemodynamics." JAMA Cardiol. 2025 Jan 1;10(1):98-100. http://dx.doi.org/10.1001/jamacardio.2024.4596 ; PMID: 39549275; PMCID: PMC11569412. , Jan-2025
Articles in Peer-reviewed Journals	Hedge ET, Brazile TL, Hughson RL, Levine BD. "Plasticity of the heart in response to changes in physical activity." J Physiol. 2024 Aug 20. Epub ahead of print. https://doi.org/10.1113/JP284158 ; PMID: 39162309 , Aug-2024
Articles in Peer-reviewed Journals	Xu J, Haigney MC, Levine BD, Dineen EH. "The tactical athlete: Definitions, cardiovascular assessment, and management, and "Fit for Duty" Standards." Cardiol Clin. 2023 Feb;41(1):93-105. Review. https://doi.org/10.1016/j.ccl.2022.08.008 ; PMID: 36368814 , Feb-2023
Articles in Peer-reviewed Journals	Thosar SS, Taqui S, Davidson B, Belcik T, Hodovan J, Rice SPM, Lindner JR. "Resting coronary flow drives the daily pattern in coronary flow reserve in patients with chest pain without obstructive epicardial stenosis." Front Cardiovasc Med. 2023 Jan 25;10:1057692. https://doi.org/10.3389/fcvm.2023.1057692 ; PMID: 36760564; PMCID: PMC9905716 , Jan-2023
Conference Materials (Downloadable)	Chauhan S, Bungo MW, Lindner JR, Koneru S, Budoff MJ, Ehrenborg KS, Macias BR, Martin DS, Levine BD. "Coronary anatomy and physiology during 1 year in space with assessment of the risk of developing atrial fibrillation. " 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-30, 2025. , Jan-2025 HRP Poster 2025.ppt